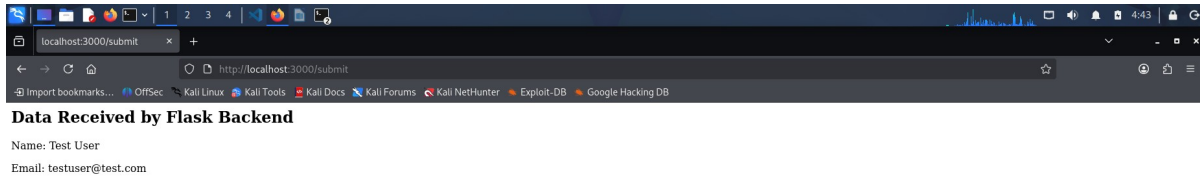


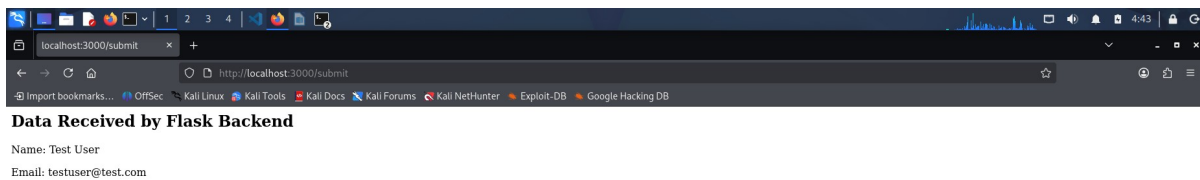
1. Frontend (Node.js with Express):

- Create a frontend using Express and Node.js.
- Include a form similar to the one from the Flask Assignment 2.
- Configure the form to send a request to the Flask backend.

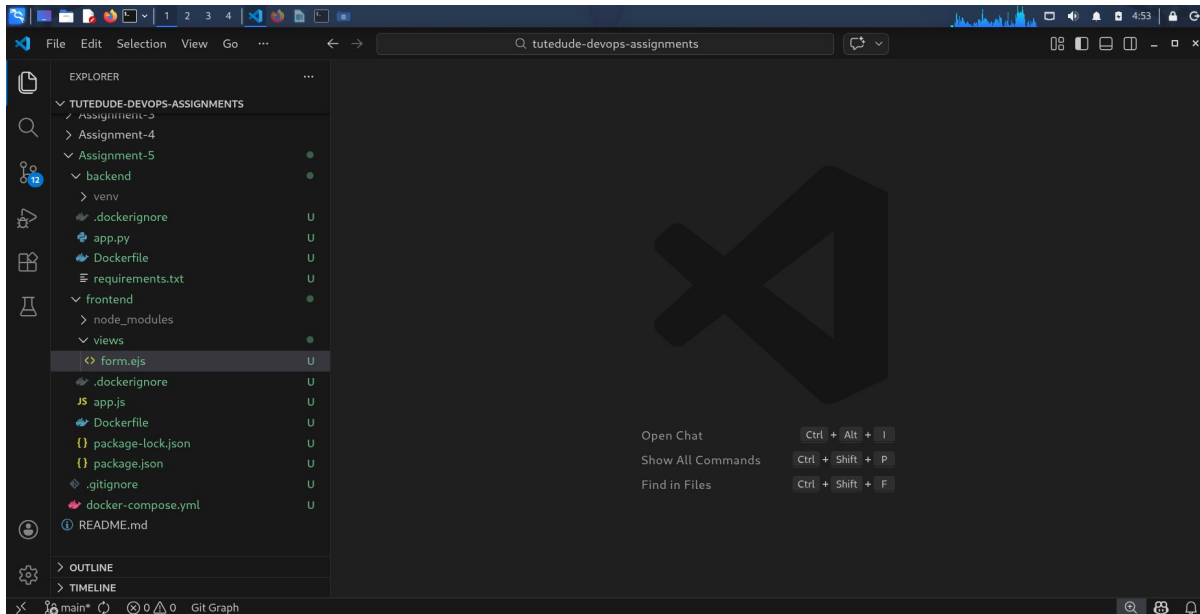


Backend (Flask):

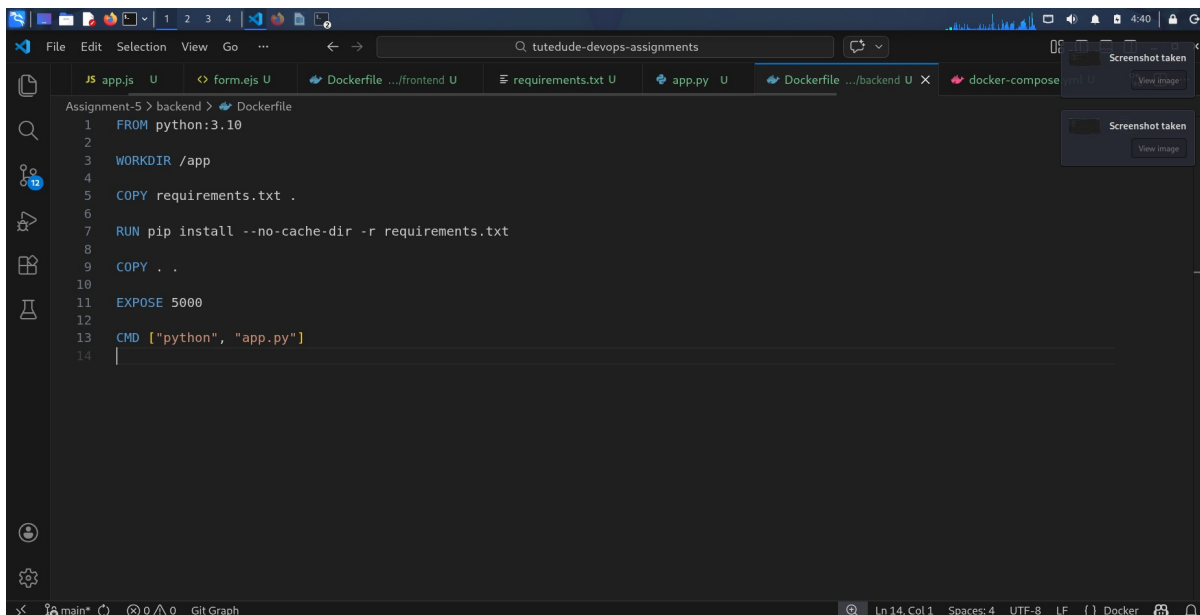
- Use the Flask backend to handle the form submission and process the data.

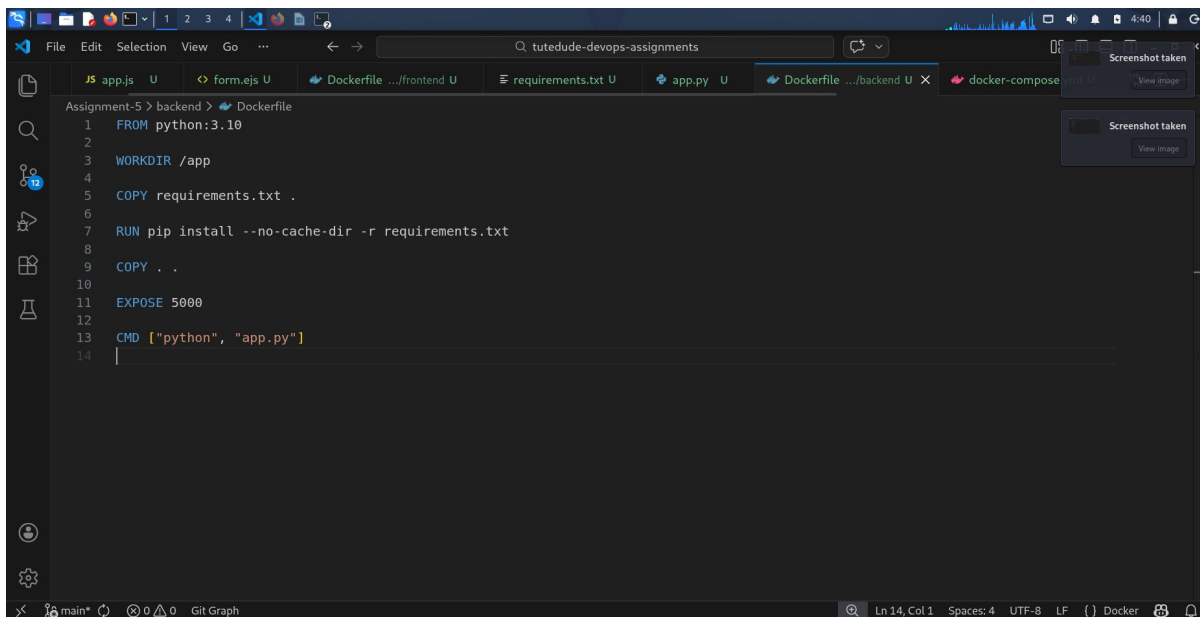


- **Folder Structure:**
- Organize the project with separate folders for the frontend and backend.



- **Docker Configuration:**
- Create a `Dockerfile` for both the frontend and backend.



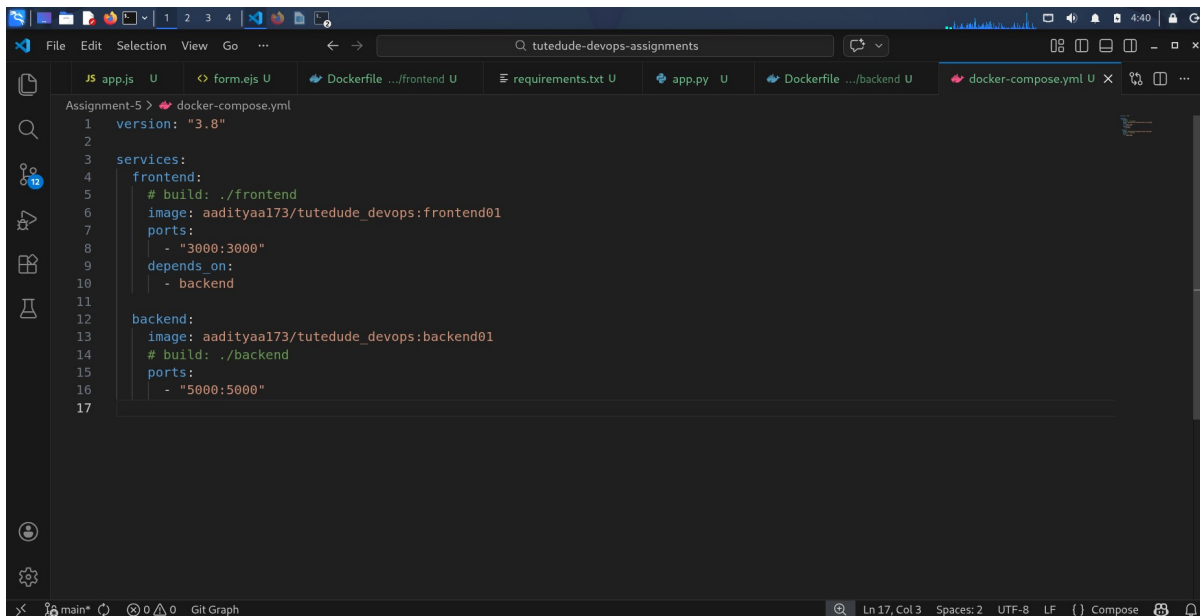


The screenshot shows a VS Code editor window with the file explorer on the left and the Dockerfile editor open. The Dockerfile contains the following content:

```
Assignment-5 > backend > Dockerfile
1 FROM python:3.10
2
3 WORKDIR /app
4
5 COPY requirements.txt .
6
7 RUN pip install --no-cache-dir -r requirements.txt
8
9 COPY . .
10
11 EXPOSE 5000
12
13 CMD ["python", "app.py"]
14
```

The status bar at the bottom indicates the file is named 'Dockerfile' and is located in the 'backend' directory.

- Write a `.yaml` file (Docker Compose) to connect both services in the same network.

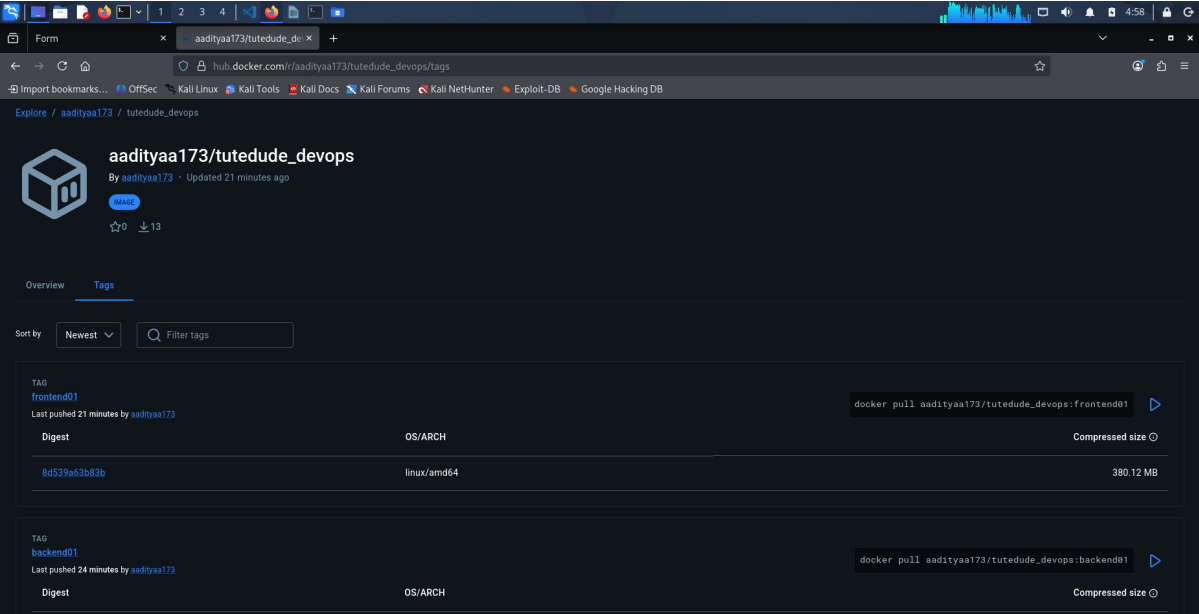


The screenshot shows a VS Code editor window with the file explorer on the left and the docker-compose.yml editor open. The docker-compose.yml file contains the following content:

```
Assignment-5 > docker-compose.yml
1 version: "3.8"
2
3 services:
4   frontend:
5     # build: ./frontend
6     image: aadityaa173/tutedude_devops:frontend01
7     ports:
8       - "3000:3000"
9     depends_on:
10      - backend
11
12   backend:
13     image: aadityaa173/tutedude_devops:backend01
14     # build: ./backend
15     ports:
16       - "5000:5000"
17
```

The status bar at the bottom indicates the file is named 'docker-compose.yml' and is located in the 'Assignment-5' directory.

- Upload both images to docker hub and push your whole code to github and add the node_modules and other non required files(.vscode) in .gitignore



The screenshot shows the Docker Hub repository page for `aadityaa173/tutedude_devops`. The repository is owned by `aadityaa173` and was updated 21 minutes ago. It has 0 stars and 13 downloads. The page is divided into two tabs: `Overview` and `Tags`, with `Tags` being the active tab. Below the tabs, there is a 'Sort by' dropdown set to 'Newest' and a 'Filter tags' input field. The main content area displays two tags: `frontend01` and `backend01`. Each tag entry includes the tag name, the time it was last pushed (21 minutes for frontend01 and 24 minutes for backend01), a 'docker pull' command, and a table with columns for 'Digest', 'OS/ARCH', and 'Compressed size'. For `frontend01`, the digest is `8d539a63b83b`, the OS/ARCH is `linux/amd64`, and the compressed size is `380.12 MB`. For `backend01`, the digest is `8d539a63b83b`, the OS/ARCH is `linux/amd64`, and the compressed size is `380.12 MB`.

Tag	Last pushed	Digest	OS/ARCH	Compressed size
<code>frontend01</code>	21 minutes by <code>aadityaa173</code>	<code>8d539a63b83b</code>	<code>linux/amd64</code>	380.12 MB
<code>backend01</code>	24 minutes by <code>aadityaa173</code>	<code>8d539a63b83b</code>	<code>linux/amd64</code>	380.12 MB